

**DEPARTMENT POLICY ON INVESTIGATIONS FOR CHILDREN PRESENTED WITH SEIZURES TO
PAEDIATRIC EMERGENCY DEPARTMENT HOSPITAL TUNKU AZIZAH (PEDHTA)**

1.0 BACKGROUND

Seizures account for 1% of all emergency department visits for children, and the aetiologies range from benign to life-threatening(1). Preliminary data in 2023 has shown that 92% of patients with seizures presenting to PEDHTA were subjected to blood taking for serum calcium, magnesium and phosphate as part of their routine investigations. None of them had significant electrolyte abnormalities or require emergent intervention to correct any electrolyte derangement. Further analysis was conducted to find out the reasons for this practice. Besides not having a standard protocol on how to appropriately investigate these patients, the most common reason reported was due to instructions by the primary team. This shows no uniformity in the practice among clinicians, not only in the emergency department but also among our paediatric counterparts.

The challenge for emergency clinicians is to balance diagnosing and treating the life threatening causes of seizures while avoiding unnecessary radiation exposure and painful procedures in patients who are unlikely to have an emergent pathology. When making decisions regarding laboratory testing, the treating physician must weigh costs, delays in patient care, and subjecting the patient to painful procedures against the likelihood of uncovering an unsuspected clinically significant abnormality. Due to paucity of data and conflicting evidence on this matter, there is no standard protocol to recommend what is the best options in terms of serum electrolytes (serum calcium, magnesium or phosphate) measurement for these children.

Previous works have demonstrated a very low incidence of clinically significant serum chemistry abnormalities for children following a febrile seizures(2,3). The American Academy of Paediatrics' practice parameter states that these studies need "not be performed routinely" for children following a first simple febrile seizures(2). On the other hand, guidelines for testing following a nonfebrile seizures are less absolute. Some do not recommend routine serum chemistry analysis in this setting while others recommend routine testing or have found a high incidence of abnormalities (2).

A study in US (2) found out that hypocalcaemia was found more often in young infants as compared to older children with non-febrile seizures in their emergency department (ED) even without pre-existing metabolic disorder. The study also found similar abnormalities in infants actively seizing in the ED or presented as status epilepticus.

Few studies (4-6) have recommended the attending physician to perform a thorough history and detailed clinical examination before deciding on the set of lab investigations. Evidence showed that the patients with certain comorbidities predispose them to develop significant electrolyte derangements that can cause seizures. These group pf patients will be benefited from measurement of serum calcium and magnesium in the ED (7). This include metabolic disorders, gastrointestinal diseases, renal insufficiency and previous history of hypocalcaemia and/or hypomagnesemia.

2.0 OBJECTIVES

- i To provide clear indications and avoid unnecessary investigations for patients presenting with seizures to PEDHTA
- ii To standardize the practice of ordering investigation for children with seizures across the institution.

3.0 SCOPE OF PRACTICE

This policy will be applied to all paediatric patients presenting to PEDHTA with active or history of seizures.

4.0 CLINICAL DECISION TOOL

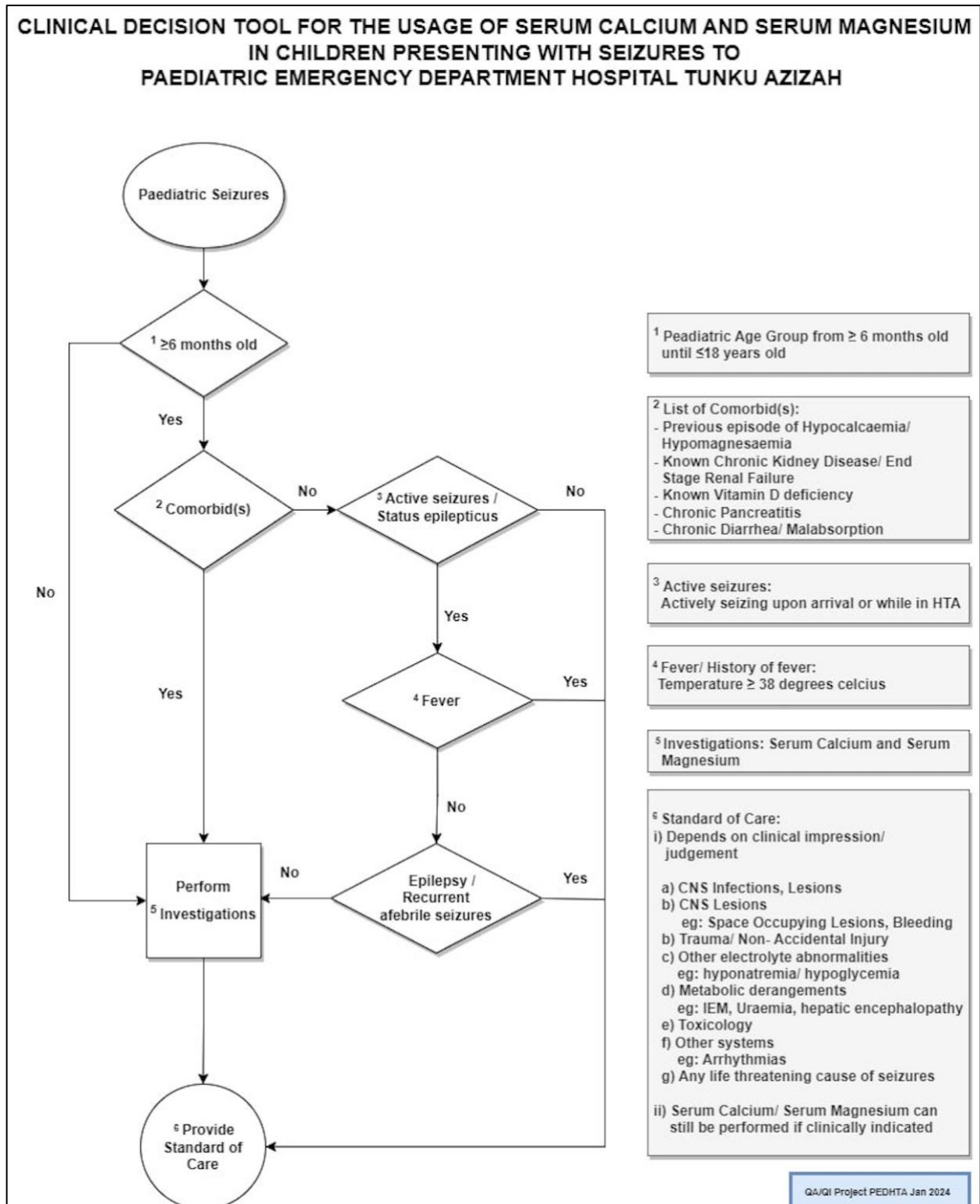
For paediatric patients presenting to PEDHTA with seizures, clinical evaluation and resuscitation of patients take priority using a standardise life support protocol. Clinicians shall also institute treatment aiming at stopping the seizures activity if the child is actively seizing at time of presentation.

Following initial stabilisation, the attending clinician will decide on the need to investigate based on thorough history and physical examination. We have developed a clinical decision tool to help the clinician to decide on the appropriate investigations (serum calcium and magnesium) to be performed (Flowchart 1).

The criteria for measuring serum electrolytes (serum calcium and magnesium) in patients presenting with seizures to PEDHTA are as follows:

- 1) Infant age < 6 months
- 2) Actively seizing or patient with status epilepticus
- 3) Patients with underlying co-morbidities that predisposes to calcium & magnesium derangements:
 - i. Previous history of hypocalcaemia or hypomagnesaemia
 - ii. Known Chronic Kidney Disease (CKD) or End Stage Renal Failure (ESRF)
 - iii. Known Vitamin D Deficiency
 - iv. Chronic Pancreatitis
 - v. Chronic Diarrhoea or malabsorption

Flowchart 1



5.0 REFERENCES

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